

Uncertainty and failure analysis

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Transfer learning

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Embeddings and neighbourhoods

Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Training a small image model

Without opening the old lesson, explain one key idea from Training a small image model. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Count both types of error for Uncertainty and failure analysis. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Uncertainty and failure analysis. Proposed explanation: “A low p_1 always means low confidence”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Analyse new outputs independently

Four new cases have (p_1 , reference label): K (0.85, 0), L (0.70, 1), M (0.30, 0), N (0.05, 0). Give predictions and confidences, overall accuracy, and accepted cases at cutoff 0.80. Calculate coverage and accepted-case accuracy. Does this policy improve accuracy on the accepted subset here?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Separate confidence from correctness

For B, E and F, give the predicted class, selected-class confidence and correctness. Explain why $p_1 = 0.10$ does not mean F has low confidence in its predicted class.

My reasoning and evidence

6. W2 Count both types of error

For all six cases, list false positives and false negatives, count true positives and true negatives, and calculate overall accuracy. Why might a single accuracy hide useful information?

My reasoning and evidence

7. W3 Audit the review policy

At cutoff 0.80, list accepted and deferred cases, coverage and accepted-case accuracy. Can you calculate final accuracy after human review from the supplied data? What does the report function return for accepted-case accuracy at cutoff 0.99?

My reasoning and evidence

8. W4 Analyse new outputs independently

Four new cases have (p1, reference label): K (0.85, 0), L (0.70, 1), M (0.30, 0), N (0.05, 0). Give predictions and confidences, overall accuracy, and accepted cases at cutoff 0.80. Calculate coverage and accepted-case accuracy. Does this policy improve accuracy on the accepted subset here?

My reasoning and evidence

9. W5 Design a failure investigation

All three blurred cases are wrong. A team claims blur caused the errors and plans to put clear images in training and their blurred copies in final evaluation. Identify the unsupported claim and the split problem; propose a better investigation.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled uncertainty and failure analysis evidence sheet to Image-pattern Network Investigation. Show how you can distinguish a predicted class, confidence score and observed correctness. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Uncertainty and failure analysis

Keep positive-class probability separate from confidence in the chosen label.

Supplied $p(\text{class } 1)$: A: 0.9 reference 0; B: 0.6 reference 1; C: 0.2 reference 0. Predict class 1 at $p \geq 0.5$; $\text{confidence} = \max(p, 1-p)$.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Positive probability is not always chosen-label confidence

	p(class 1)	prediction	confidence	reference
A	0.9	1	0.9	0
B	0.6	1	0.6	1
C	0.2	0	0.8	0

Read a cell using both its row and column labels.

Uncertainty and failure analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predictions

A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct.



2. Selected-label confidence

Confidences are 0.9, 0.6, 0.8. C's confidence is 0.8 despite $p(\text{class } 1)=0.2$.



3. Selective review

Accept confidence ≥ 0.8 : A and C; accuracy 1/2 and coverage 2/3.

CHANGE ONE THING

Raise the acceptance cutoff to 0.85.

Explain the changed result using the matching lesson.

What is C's positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep positive-class probability separate from confidence in the chosen label.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Accept confidence ≥ 0.8 : A and C; accuracy $1/2$ and coverage $2/3$.

1. Selected-label confidence
2. Selective review
3. Predictions

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Selective review Accept confidence ≥ 0.8 : A and C; accuracy $1/2$ and coverage $2/3$. Positive-class probability and confidence in the chosen label differ when the predicted class is zero. For C, a class-one probability of 0.2 gives class-zero confidence 0.8 under this toy rule. Selective accuracy counts only accepted cases; coverage counts how many were accepted. A stricter cutoff can retain a confidently wrong case and lower accuracy on this supplied set.

Practice 2: explain the deciding evidence

Changed-case practice: Raise the acceptance cutoff to 0.85. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. Only A is accepted: accuracy $0/1$, coverage $1/3$.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Only A is accepted: accuracy 0/1, coverage 1/3. The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

Repair a missing building block

Review Transfer learning (ai-5/28). Goal: Trace a reused feature extractor and a newly fitted output head. Short practice: Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Embeddings and neighbourhoods (ai-5/26). Goal: Look up a vector using a token ID and preserve sequence shape. Short practice: What would justify claiming nearby vectors mean related topics? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Training a small image model (ai-5/23). Goal: Trace a complete image model with fixed features and one trained weight. Short practice: Does lower training loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
